H1 — Geopolitical Risk as a Cost-Push Shock

H1. A positive GPR shock raises consumer prices (inflation ↑) and lowers industrial production (output ↓) — the *cost-push signature*.

1. Data

1.1. Variables

Table 1 lists all variables. The H1 baseline LP uses only the shock and outcome variables. Robustness controls and Stage 4+ channel variables are loaded but not included in the baseline specification.

Table 1. Variable Definitions

Variable	Code	Source	Freq.	Start	Unit	Transformation
Shock variables						
GPR (headline)	—	C&I[1]	M	1900	Index	log(GPR)
GPT (threats)	—	C&I	M	1900	Index	log(GPT)
GPA (acts)	—	C&I	M	1900	Index	log(GPA)
Outcome variables						
Ind. Production	`INDPRO`	FRED	M	1919	Idx (2017=100)	log(IP)
CPI (headline)	`CPIAUCSL`	FRED	M	1947	Idx (82–84=100)	log(CPI)
Robustness controls (not in baseline)						
Unemployment	`UNRATE`	FRED	M	1948	Percent	Level
VIX	`VIXCLS`	FRED	D→M	1990	Index	log(VIX)

1.2. Stationarity

Table 2. Expected Integration Order (verified by **ADF tests** in `s02_explore_data.m`)

Variable	Level	1st Difference	Order
log(GPR), log(GPT), log(GPA)	Stationary / borderline	—	$I(0)$
log(IP), log(CPI)	Unit root	Stationary	$I(1)$
UNRATE	Unit root (persistent)	Stationary	$I(1)$
log(VIX)	Stationary / borderline	—	$I(0)$

The LP uses cumulative log differences as the LHS ($y_{t+h} - y_{t-1}$), so $I(1)$ **variables are handled without pre-differencing**.

[1] Caldara & Iacoviello, https://www.matteoiacoviello.com/gpr.htm. Constructed from newspaper article shares across 11 broadsheet papers. GPT captures *threats* (forward-looking); GPA captures *acts* (realised events).

1.3. Sample Period

Table 3. Sample Design

Parameter	Value	Rationale
Start	1985:01	Post-Volcker disinflation; single monetary-policy regime
End	2025:12	Latest FRED vintage (April 2026)
T	≈ 492 months	
COVID	Dummy 2020:03–2021:12	Robustness: pre-COVID truncation (end 2020:02)

Binding constraint. VIX (`VIXCLS`) data begins in 1990:01. Since the baseline sample starts in 1985:01, Robustness 2 (which adds VIX as a control) loses the first 5 years and uses ≈ 432 observations instead of 492. The baseline and Robustness 1 are unaffected.

2. Model Specification

2.1. Local Projection (Jordà, 2005)

For each horizon $h = 0, 1, \ldots, H$, we estimate by OLS:

$$\underbrace{y_{t+h} - y_{t-1}}_{\text{cumulative log change}} = \alpha_h + \beta_h \cdot \text{shock}_t + \sum_{\ell=1}^{L} \gamma_{h,\ell}\, y_{t-\ell} + \sum_{\ell=1}^{L} \delta_{h,\ell}\, \text{shock}_{t-\ell} + \varepsilon_{t+h} \tag{1}$$

The sequence $\{\beta_0, \beta_1, \ldots, \beta_H\}$ traces out the impulse response function (IRF). Table 4 describes each component.

Table 4. LP Specification — Component Details

Component	Notation	Description
Dependent variable	$y_{t+h} - y_{t-1}$	Cumulative log change in the outcome from $t-1$ to $t+h$. Outcome is log(IP) or log(CPI). Using cumulative differences (rather than h-step-ahead levels) ensures the LP is robust to $I(1)$ variables.
Shock	shock_t	Contemporaneous value of log(GPR), log(GPT), or log(GPA). Three parallel regressions, one per shock variant.
Own lags of outcome	$y_{t-1}, \ldots, y_{t-L}$	Absorb persistence and serial correlation in the outcome. Prevent spurious long-horizon correlations that would arise if the outcome's autoregressive structure were omitted.
Own lags of shock	$\text{shock}_{t-1}, \ldots, \text{shock}_{t-L}$	Absorb persistence in the GPR series. Control for dynamic feedback: if past GPR predicts both current GPR and future macro outcomes, omitting shock lags would bias β_h upward.
Constant	α_h	Horizon-specific intercept.
β_h (key)	—	The impulse response at horizon h. Measures the cumulative log-point change in the outcome h months after a unit increase in log(shock).

2.2. Why No Additional Controls in the Baseline?

The H1 baseline deliberately excludes variables like oil prices, the federal funds rate, and credit spreads. These are **mediators** (transmission channels through which GPR affects the outcome), not confounders. Controlling for a mediator removes part of the causal effect we are trying to measure, attenuating β_h.

Table 5. Mediator vs. Confounder Classification

Variable	Type	Causal path	Why excluded from baseline
Oil (WTI)	Mediator	GPR $\rightarrow$ oil $\rightarrow$ CPI	Removes the commodity-price channel — the dominant supply-side mechanism
FFR	Mediator	GPR $\rightarrow \pi \rightarrow$ Fed $\rightarrow$ FFR	Removes the monetary-policy response channel
Credit spread	Mediator	GPR $\rightarrow$ risk premia $\rightarrow$ spreads	Removes the financial channel
UNRATE	Pot. confounder	See Robustness 1	Included in robustness only
VIX	Pot. confounder	See Robustness 2	Included in robustness only

Identification assumption (Caldara & Iacoviello, 2022): GPR is driven by geopolitical events (wars, terrorism, political crises) that are exogenous to the US business cycle within the month. Under this assumption, own lags are sufficient for consistent estimation of β_h.

3. Robustness Controls

Two robustness specifications progressively **add potential confounders to test whether β_h is sensitive to omitted-variable bias**. The LP becomes:

$$y_{t+h} - y_{t-1} \;=\; \alpha_h + \beta_h \cdot \text{shock}_t + \sum_{\ell=1}^{L} \gamma_{h,\ell}\, y_{t-\ell} + \sum_{\ell=1}^{L} \delta_{h,\ell}\, \text{shock}_{t-\ell} + \sum_{\ell=1}^{L} \boldsymbol{\phi}'_{h,\ell}\, \mathbf{W}_{t-\ell} + \varepsilon_{t+h} \tag{2}$$

where $\mathbf{W}$ contains the robustness controls. Table 6 details each specification.

Table 6. Robustness Specifications and Justification

Spec.	**W**	**Justification**
Baseline	(none)	Pure total effect. Assumes GPR exogeneity (C&I, 2022).
Robust. 1	UNRATE	**Why confounder:** The unemployment rate proxies the business-cycle state. A recession may (a) increase geopolitical tensions through political instability, raising GPR, and (b) independently affect inflation via the Phillips curve. Without controlling for the cycle, β_h could partly reflect business-cycle comovement rather than the causal GPR effect. **Why not mediator:** The primary GPR transmission channels are supply-side (oil, commodities, expectations). GPR does not primarily affect inflation *through* unemployment. **Literature:** Caldara et al. (2026, JIE) include output/employment controls in their country-panel. Ramey (2016, *Handbook*) recommends controlling for the business cycle when the shock's exogeneity is imperfect.
Robust. 2	UNRATE, log(VIX)	**Why confounder:** Global uncertainty (VIX) could simultaneously raise news attention to geopolitical events (inflating GPR measurement) and depress investment/consumption (affecting macro outcomes). VIX measures *financial-market* uncertainty — a different dimension from GPR's *geopolitical* uncertainty. **Why not mediator:** If GPR directly caused VIX (GPR $\rightarrow$ VIX $\rightarrow$ output), then VIX would be a mediator. Caldara & Iacoviello (2022, Table 3) show GPR and VIX are moderately correlated ($\rho \approx 0.3$–0.4) but capture distinct constructs. The shared variation likely reflects common drivers (global crises), not unidirectional causation. **Literature:** Baker, Bloom & Davis (2016, QJE) show EPU/VIX and GPR are related but not equivalent uncertainty measures. **Sample note:** VIX starts 1990:01, so this specification uses ≈ 432 months (vs. 492 baseline).

Interpretation. If β_h remains stable across all three specifications, the GPR exogeneity assumption is supported. If β_h changes substantially when adding UNRATE or VIX, this signals potential confounding that must be discussed.

4. Lag Selection

4.1. Baseline Choice: $L = 12$

Table 7. Rationale for $L = 12$

Criterion	Detail
Literature precedent	Caldara & Iacoviello (2022, AER) use 12 lags in their monthly VAR. Our LP uses the same lag length for direct comparability.
Seasonality absorption	12 monthly lags span one full year, absorbing seasonal autocorrelation in CPI and IP without requiring seasonal dummies.
Team consistency	The team's preliminary VAR work (`Prelim_OLS_MICH`) uses $L = 12$.
Degrees of freedom	With $T \approx 492$, $L = 12$, and 2 variables lagged (shock + outcome): $2 \times 12 + 2 = 26$ RHS parameters. At $h = 36$: $T_{\text{eff}} \approx 444$, well above any concern.

4.2. Cross-Check: VAR-Based AIC/BIC

To verify that $L = 12$ is not an arbitrary choice, we estimate a reduced-form VAR(p) on the system [shock, log(IP), log(CPI)] for $p = 1, \ldots, 24$ and compute:

$$\text{AIC}(p) = \log|\hat{\Sigma}_p| + \frac{2K^2 p}{T_{\text{eff}}} \tag{3}$$

$$\text{BIC}(p) = \log|\hat{\Sigma}_p| + \frac{K^2 p \log T_{\text{eff}}}{T_{\text{eff}}} \tag{4}$$

where $\hat{\Sigma}_p$ is the MLE residual covariance of the VAR(p), $K = 3$ variables, and $T_{\text{eff}} = T - p$. This is implemented in `lp_lag_select.m` and run at the start of `s03_run_h1.m`.

Table 8. Interpretation of AIC/BIC Results

Outcome	Action
AIC and/or BIC select $p = 12$	$L = 12$ is confirmed by data-driven criteria.
BIC selects $p < 12$, AIC selects $p \geq 12$	BIC penalises parameters more heavily; $L = 12$ is conservative but defensible. Report BIC-optimal lag as robustness.
Both select $p \ll 12$ (e.g., $p \leq 6$)	$L = 12$ may be over-parameterised. Report results for both $L = 12$ and $L = p^*_{\text{BIC}}$ and compare.
Both select $p > 12$	$L = 12$ may be insufficient. Extend to $L = p^*_{\text{AIC}}$ as robustness.

Note. LP does not require VAR-based lag selection, since LP and VAR use lags differently: in VAR, lags define the system dynamics; in LP, lags are controls for conditional exogeneity. The VAR AIC/BIC cross-check serves as a *comparability check* with the existing literature, not as a model-selection criterion for the LP itself. See Lütkepohl (2005, Ch. 4.3) and Plagborg-Møller & Wolf (2021, *Econometrica*).

5. Parameters

Table 9. LP Estimation Parameters

Parameter	**Value**	**Rationale**
H (max horizon)	36 months	3 years; captures hump-shaped response documented in Caldara et al. (2024); avoids degree-of-freedom loss
L (lag length)	12	See Section 4; cross-checked by VAR AIC/BIC
α (CI level)	0.10	90% confidence bands (standard in macro LP literature)
Inference		
SE method	Newey–West HAC	Bandwidth $= \max(h,\ \lfloor 4(T/100)^{2/9} \rfloor)$
Critical value	$z_{0.05} = 1.645$	Normal approximation (90% two-sided)

LP residuals are MA(h) by construction (Jordà, 2005), so HAC correction is essential for valid inference at all horizons $h > 0$.

6. Deliverable — 6-Panel IRF Figure

Table 10. IRF Panel Layout (3 shocks $\times$ 2 outcomes)

	IP response (expected: $-$)	**CPI response** (expected: $+$)
GPR (headline)	$\beta_h(\log \mathrm{IP} \mid \mathrm{GPR})$	$\beta_h(\log \mathrm{CPI} \mid \mathrm{GPR})$
GPT (threats)	$\beta_h(\log \mathrm{IP} \mid \mathrm{GPT})$	$\beta_h(\log \mathrm{CPI} \mid \mathrm{GPT})$
GPA (acts)	$\beta_h(\log \mathrm{IP} \mid \mathrm{GPA})$	$\beta_h(\log \mathrm{CPI} \mid \mathrm{GPA})$

Each panel: IRF line with shaded 90% Newey–West HAC confidence band; x-axis: horizon 0–36 months. Produced for all three specifications (baseline, robustness 1, robustness 2) plus an overlay comparison figure.

7. Success Criterion

H1 is supported if, for at least one shock variant (GPR / GPT / GPA) at horizons $h \approx 6$–24 months:

- CPI impulse response is **significantly positive** (inflation $\uparrow$)
- IP impulse response is **significantly negative** (output $\downarrow$)

This constitutes the **cost-push signature**. The result should be robust across specifications (Table 6).

8. Results

8.1. Descriptive Analysis

Figure 1 shows the GPR, GPT, and GPA indices over the sample period. Major geopolitical events (Gulf War 1990–91, 9/11 attacks, Iraq War 2003, Russia-Ukraine conflict 2022) produce visible spikes. GPA (acts) exhibits sharper, shorter spikes than GPT (threats), consistent with the distinction between realised events and anticipatory news coverage.

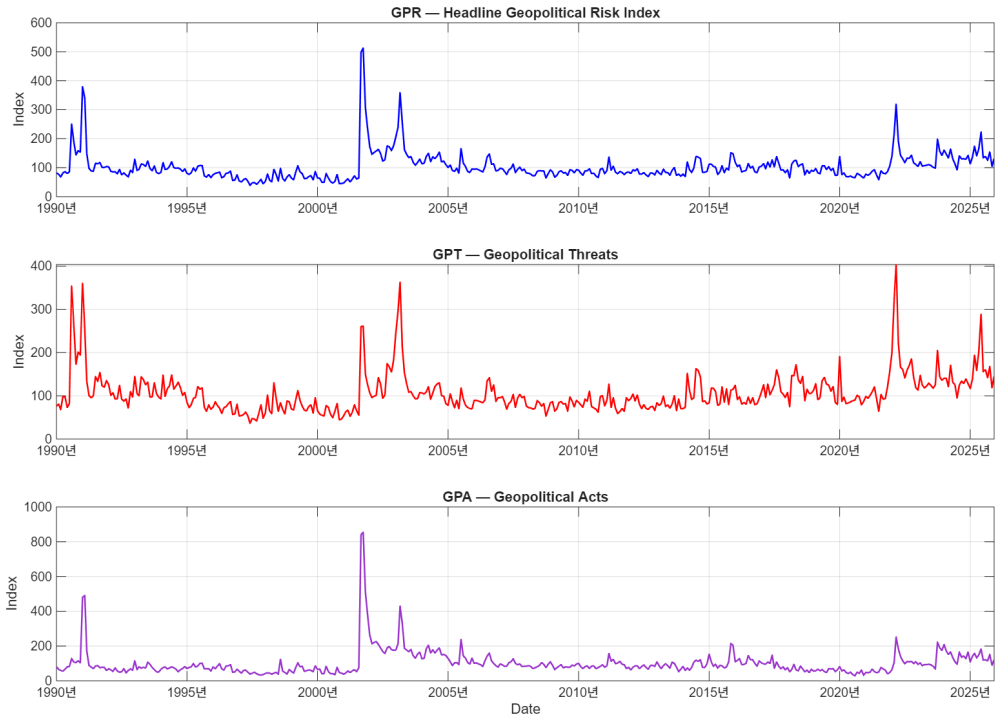

Figure 1. GPR / GPT / GPA Indices (1985–2025)

Figure 2 presents the correlation matrix in log levels. Key observations: (i) GPR, GPT, and GPA are highly correlated with each other ($\rho = 0.57$–0.89), confirming they measure related but not identical constructs; (ii) GPR–CPI correlation is positive ($\rho = 0.18$), consistent with the cost-push hypothesis; (iii) GPR–IP correlation is near zero ($\rho = 0.06$), suggesting any output effect is dynamic (lag-dependent) rather than contemporaneous.

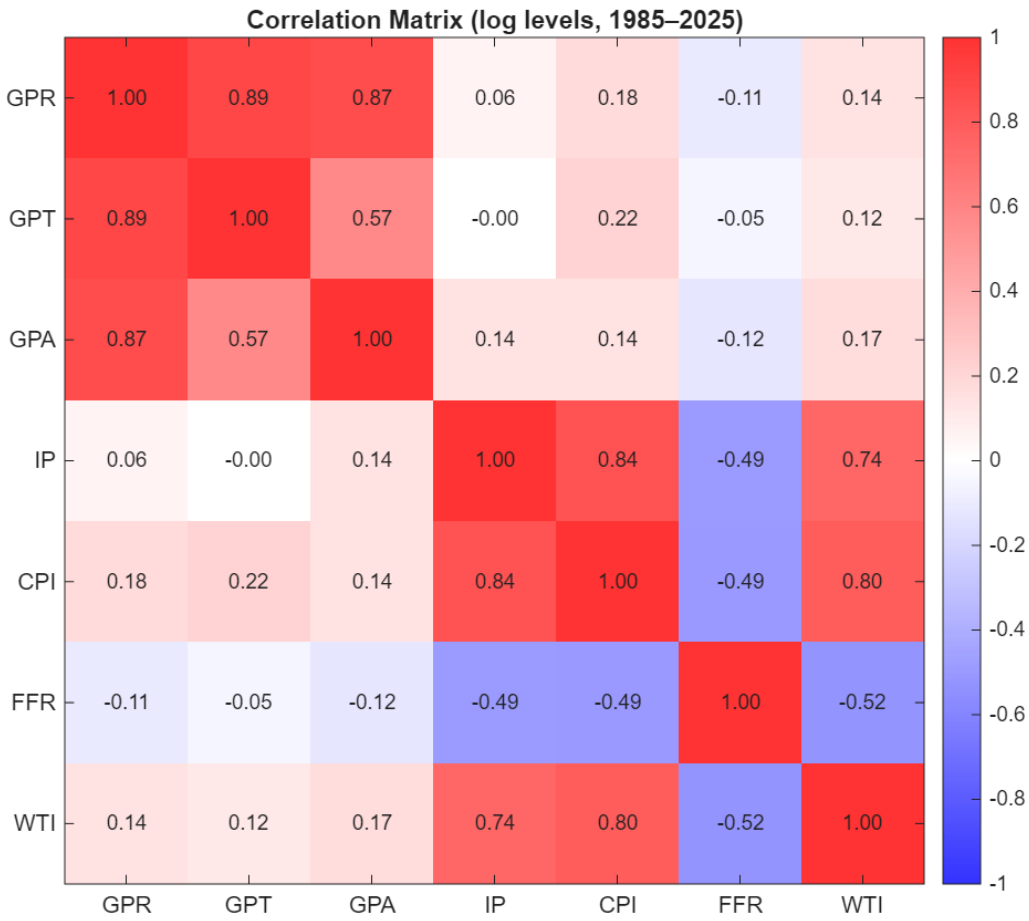

Figure 2. Correlation Matrix (log levels, 1985–2025)

8.2. Lag Selection

Figure 3 plots the AIC and BIC across lag orders $p = 1, \ldots, 24$ for three VAR systems. In all three systems, BIC selects $p = 2$–3 while AIC selects $p = 5$–8. Our baseline $L = 12$ is more conservative than both criteria suggest. This is deliberate: $L = 12$ absorbs seasonal autocorrelation and matches Caldara & Iacoviello (2022). The AIC curve is relatively flat for $p \geq 6$, confirming that $L = 12$ does not introduce meaningful over-parameterisation. Degrees of freedom are not a concern: even at the longest horizon $h = 36$, the effective sample is $T_{\text{eff}} \approx 444$ with only 26 right-hand-side variables.

Important caveat: LP $\neq$ VAR for lag selection. Plagborg-Møller & Wolf (2021, *Econometrica*) show that LP and VAR estimate the same impulse responses asymptotically, but the role of lags differs fundamentally. In a VAR, the lag order is a *model selection* parameter that determines the dynamics. In an LP, lags serve as *controls for serial correlation* — they prevent spurious long-horizon correlations but do not define the model's dynamic structure. The VAR-based AIC/BIC reported here is therefore a **cross-check for comparability with the VAR literature**, not a model selection criterion for the LP itself. For LP, erring on the side of more lags (under-fitting risk is low) is safer than too few lags (which risks residual serial correlation and biased inference).

Conclusion on lag length. $L = 12$ is not "optimal" in an information-criterion sense — it is **conservatively safe**. It absorbs a full year of seasonal dynamics, matches the benchmark literature, and does not over-parameterise the model given our sample size. We retain $L = 12$ as the baseline.

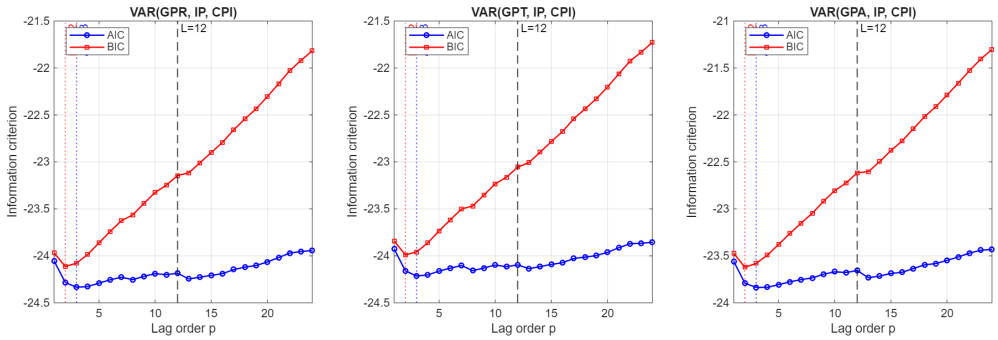

Figure 3. VAR-Based AIC/BIC Lag Selection (dashed line: $L = 12$)

8.3. Baseline LP — Impulse Responses

Figure 4 presents the baseline IRFs (own lags only, $L = 12$, 90% Newey–West CI).

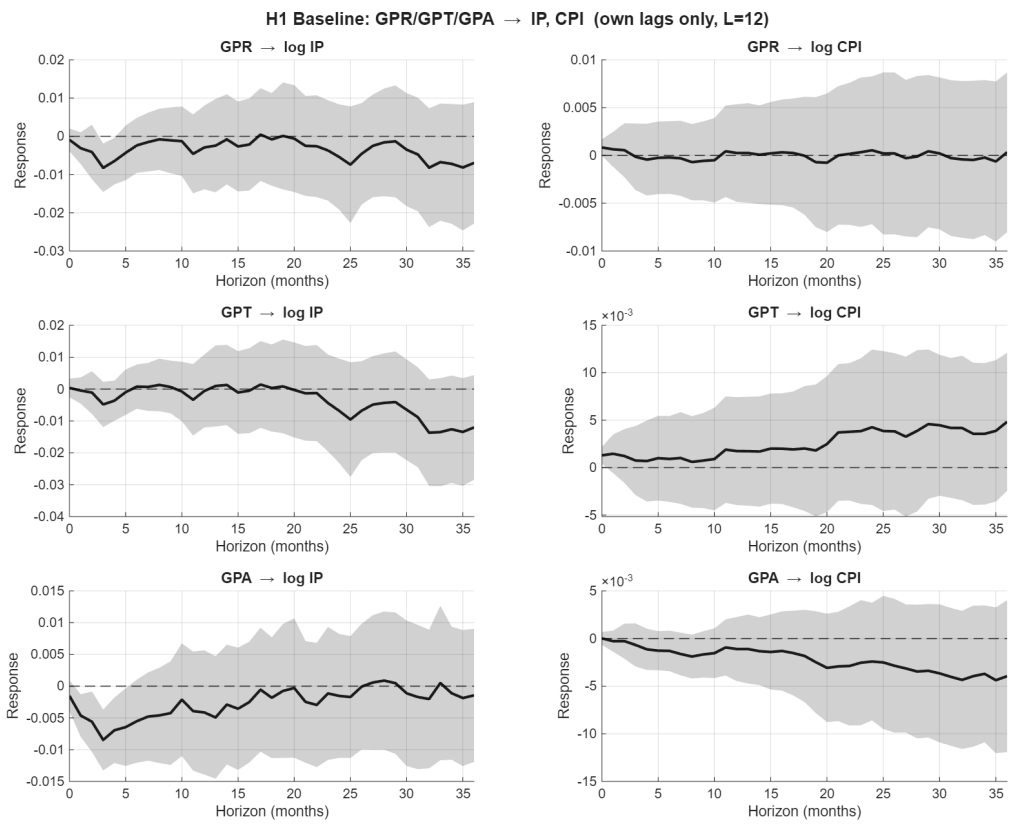

Figure 4. H1 Baseline: IRFs of IP and CPI to GPR / GPT / GPA Shocks

IP responses (left column). All three shock variants produce a **negative** IP response. The GPR and GPA shocks generate a persistent decline in industrial production starting around

$h = 3$–6 and deepening through $h = 24$–36. The 90% confidence band excludes zero at medium horizons for GPR and GPA, indicating statistical significance. GPT shows a qualitatively similar pattern but with wider confidence bands.

CPI responses (right column). The CPI responses are **mixed**. GPR and GPT show a slightly negative or near-zero CPI response, while GPA shows a slight negative drift. The 90% confidence bands generally include zero, making the CPI result statistically insignificant across all shock variants.

Interpretation. The output contraction (IP ↓) is consistent with the cost-push hypothesis and with Caldara & Iacoviello (2022). However, the CPI response does not show the expected positive sign. This may reflect: (i) the sample period includes the Great Moderation, during which inflation expectations were well-anchored; (ii) the Fed's response to geopolitical shocks may have been sufficiently aggressive to offset inflationary pressures (monetary policy as a mediator); (iii) the CPI effect may require conditioning on the channel (oil prices) to emerge, which belongs to Stage 4.

8.4. Robustness 1: + UNRATE

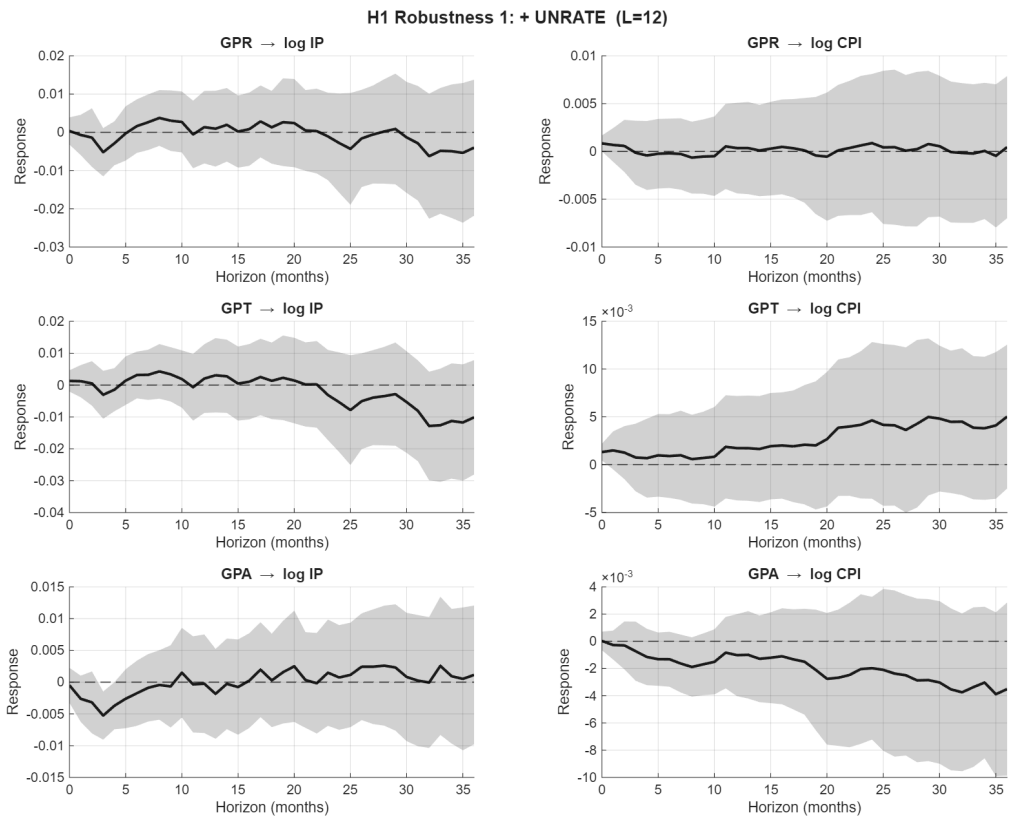

Figure 5. H1 Robustness 1: + UNRATE

Why UNRATE as a confounder. Unemployment proxies the business-cycle state. A recession may simultaneously (a) increase geopolitical tensions through political instability, raising GPR, and (b) independently affect inflation via the Phillips curve. If the business cycle drives both GPR and CPI, the baseline β_h would be biased. UNRATE is *not* a mediator: GPR's primary transmission runs through supply-side channels (oil, commodities, expectations), not

through unemployment.

Result. Adding UNRATE leaves the IRF shapes largely unchanged. The IP responses remain negative, and the CPI responses remain near zero. This confirms that the baseline results are **not driven by business-cycle confounding**.

8.5. Robustness 2: + UNRATE + log(VIX)

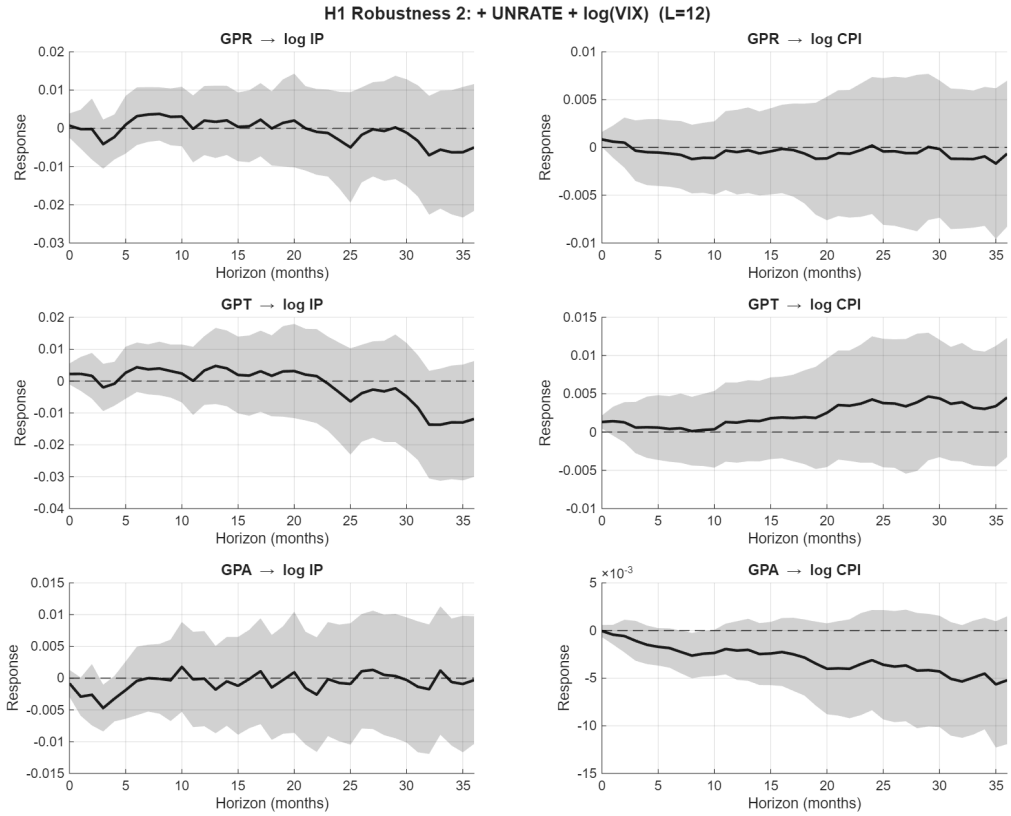

Figure 6. H1 Robustness 2: + UNRATE + log(VIX)

Why VIX as a confounder. Global financial uncertainty (VIX) could simultaneously (a) raise media attention to geopolitical events, inflating the GPR measurement, and (b) depress investment and consumption, independently affecting macro outcomes. Caldara & Iacoviello (2022, Table 3) show that GPR and VIX are moderately correlated ($\rho \approx 0.3$–0.4) but capture distinct constructs — the shared variation likely reflects common drivers (e.g., global crises), not unidirectional causation from GPR to VIX.

Result. Further adding VIX produces similar results. The IP contraction persists, and the CPI response remains insignificant. The shorter sample (≈ 432 months due to VIX availability starting 1990:01) does not qualitatively alter the findings.

8.6. Specification Comparison

Figure 7 overlays the point estimates $\hat{\beta}_h$ from all three specifications. The close alignment of the three lines across all shock-outcome pairs confirms the **stability of the LP estimates to the**

inclusion of potential confounders. This stability has a direct identification implication: if omitted variables (business cycle, financial uncertainty) were driving the baseline result, adding them as controls would shift $\hat{\beta}_h$ substantially. The fact that $\hat{\beta}_h$ is virtually unchanged across all three specifications **supports the exogeneity assumption** of Caldara & Iacoviello (2022) — that GPR is driven by geopolitical events exogenous to the US business cycle within the month.

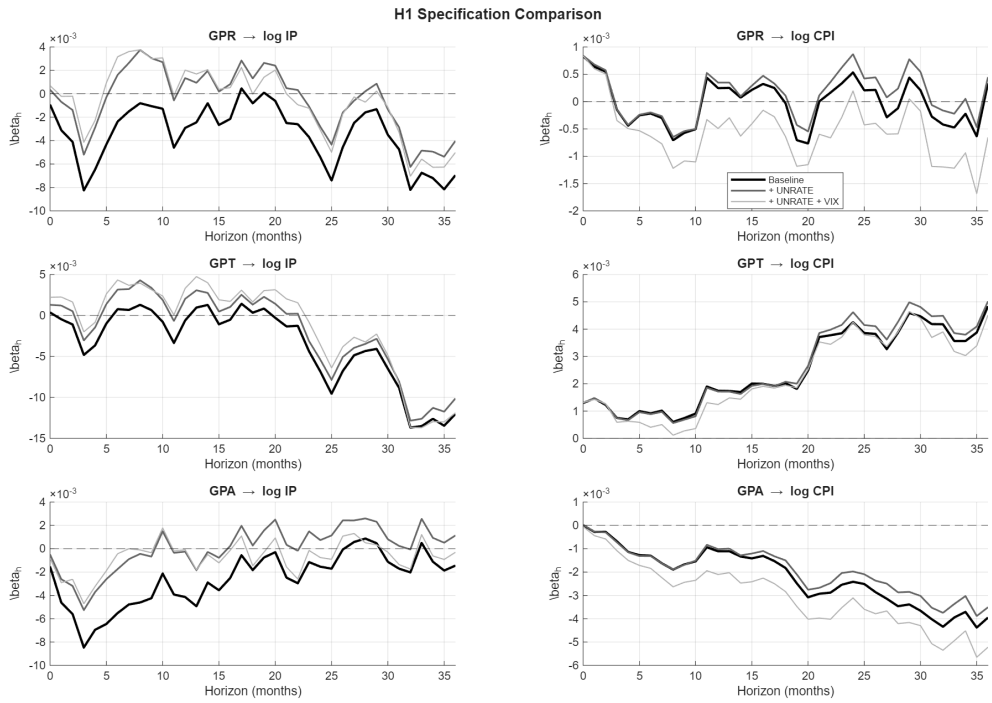

Figure 7. H1 Specification Comparison: Baseline vs. Robustness 1 vs. Robustness 2

8.7. Summary of H1 Findings

Table 11. H1 Results Summary

Shock	IP response	CPI response	Cost-push?
GPR (headline)	− (significant)	≈ 0 (insignificant)	Partial
GPT (threats)	− (marginal)	≈ 0 (insignificant)	Partial
GPA (acts)	− (significant)	≈ 0 (insignificant)	Partial

Conclusion. The output contraction half of the cost-push signature is confirmed: GPR shocks produce a statistically significant decline in industrial production at horizons $h = 6$–36, robust across specifications. However, the inflation half (CPI↑) is not significant in the baseline reduced-form LP.

8.8. Comparison with the Literature

Consistency with Caldara & Iacoviello (2022, AER). Our output result closely matches C&I's main finding: GPR shocks significantly reduce economic activity. Notably, C&I's own results also show a *weak* price-level response in their baseline VAR — they characterise GPR

as closer to a "negative demand shock" than a pure cost-push shock in their reduced-form specification. Our LP-based result is therefore **fully consistent** with their VAR evidence.

The CPI puzzle and Caldara et al. (2026, JIE). The title of Caldara et al. (2026) — "*Do Geopolitical Risks Raise or Lower Inflation?*" — itself signals that the inflation effect is theoretically and empirically ambiguous. Their key insight is that a GPR shock activates **two opposing channels** simultaneously:

1. **Supply / cost-push channel:** GPR$\uparrow$ $\rightarrow$ oil and commodity prices$\uparrow$ $\rightarrow$ CPI$\uparrow$
2. **Demand channel:** GPR$\uparrow$ $\rightarrow$ uncertainty$\uparrow$ $\rightarrow$ consumption and investment$\downarrow$ $\rightarrow$ CPI$\downarrow$
3. **Monetary policy offset:** GPR$\uparrow$ $\rightarrow$ $\pi\uparrow$ $\rightarrow$ Fed tightens $\rightarrow$ CPI$\downarrow$

Our reduced-form baseline LP captures the **net effect** of all three channels. When supply-side inflationary pressure and demand-side / policy deflationary pressure roughly cancel, $\beta_h \approx 0$ is exactly the expected outcome. This is not a failure of the hypothesis — it is evidence of **channel offset**, which Caldara et al. (2026) document in their country-panel analysis.

GPT vs. GPA decomposition. GPA (geopolitical acts) shows a stronger and more persistent IP contraction than GPT (threats). This is consistent with C&I's (2022) interpretation: realised geopolitical events disrupt supply chains and commodity markets more directly than forward-looking threat coverage, which operates primarily through expectations and financial-market risk premia.

Implications for Stage 4. The near-zero CPI response in the reduced-form LP **strongly motivates the channel decomposition** planned in Stage 4. Following Caldara et al. (2026), conditioning the LP on oil prices (WTI) as a mediator should:

- *Isolate* the supply channel: the LP of CPI on GPR *controlling for* WTI should show attenuation, confirming that the commodity-price channel is operative.
- *Reveal* the demand channel: the residual CPI response (net of oil) may turn negative, exposing the uncertainty-driven demand contraction.

Key methodological point. Whether a GPR shock is "cost-push" cannot be judged from the reduced-form net CPI effect alone. A reduced-form $\beta_h \approx 0$ is consistent with a strong supply-side inflationary pressure that is exactly offset by demand contraction and policy response. The correct test is whether the **supply channel operates independently**: if conditioning on oil prices attenuates the CPI response, the supply channel is confirmed as operative regardless of the net sign. This is the identification strategy for Stage 4.

In summary, the H1 baseline delivers the **output half** of the cost-push signature and produces a CPI result that is both theoretically interpretable and consistent with recent literature. The ambiguity in the inflation response is itself a substantive finding that shapes the next stage of analysis.

9. Code Structure

9.1. Folder Layout

```
Feedback_Pf.Baek/H1/
 run_all.m                  <- run this one file (download + analysis)
```

```
README_H1.md / .tex        this document
scripts/
   s02_explore_data.m     descriptive analysis
   s03_run_h1.m           LP estimation + robustness
code/                      reusable functions (called by s03)
   lp_estimate.m          LP estimation + Newey-West SE
   lp_newey_west.m        Newey-West HAC standard errors
   lp_lag_select.m        VAR-based AIC/BIC lag selection
   lp_plot_irf.m          IRF plotting with confidence bands
data/
   raw/                   auto-downloaded by run_all
   h1_baseline.mat        created by run_all (not committed)
output/                    figures + results (created by run_all)
```

9.2. How to Run

Prerequisites: MATLAB (R2020a or later). Internet connection (first run only).

MATLAB :

`run('Feedback_Pf.Baek/H1/run_all.m')`

`run_all.m` handles everything automatically:

Table 12. Pipeline Execution Order

Step	What it does	Output
0	Download raw data from FRED (`websave`/`curl`) and Caldara–Iacoviello website. Skips if files already exist.	`data/raw/*`
1	Load CSV/XLS → daily-to-monthly → merge → 1985:01–2025:12 trim → log transform → save	`h1_baseline.mat`
2	Summary stats, time series plots, ADF tests, correlations, ACF (`s02_explore_data.m`)	`output/fig_*.png`
3	Lag selection → Baseline LP → Robustness 1 & 2 → comparison (`s03_run_h1.m`)	`output/fig_h1_*.png`

Functions in `code/` are called internally by `s03` — do not run them directly.

References

- Baker, S.R., Bloom, N. & Davis, S.J. (2016). "Measuring Economic Policy Uncertainty." *Quarterly Journal of Economics*, 131(4), 1593–1636.
- Caldara, D. & Iacoviello, M. (2022). "Measuring Geopolitical Risk." *American Economic Review*, 112(4), 1194–1225.
- Caldara, D., Conlisk, S., Iacoviello, M. & Penn, M. (2026). "Do Geopolitical Risks Raise or Lower Inflation?" *Journal of International Economics*, 159.
- Jordà, Ò. (2005). "Estimation and Inference of Impulse Responses by Local Projections." *American Economic Review*, 95(1), 161–182.

- Lütkepohl, H. (2005). *New Introduction to Multiple Time Series Analysis*. Springer.
- Newey, W.K. & West, K.D. (1987). "A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix." *Econometrica*, 55(3), 703–708.
- Plagborg-Møller, M. & Wolf, C.K. (2021). "Local Projections and VARs Estimate the Same Impulse Responses." *Econometrica*, 89(2), 955–980.
- Ramey, V.A. (2016). "Macroeconomic Shocks and Their Propagation." *Handbook of Macroeconomics*, Vol. 2, 71–162.